SAGE: IRL Extension

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***Abstract*—Public Large Language Models (LLMs) are being increasingly explored for aiding students to learn difficult concepts in STEM education. Tutoring systems built on them can restrict what a student sees according to what they have already learned. However, such systems are usually told what a student knows rather than judging it themselves, as a topic counts as learned because a course was completed or an instructor said so. Where a system does judge, it reduces everything a student has done to a single score. A student who answers factual questions well but cannot yet do the hands-on work then looks the same as one who can do both. We call this the Evidence Diversity Problem. In this paper, we present a learner model for "SAGE" (Student-focused Adaptive Guidance Engine), an intelligent tutoring framework that enforces instructor-regulated learning. The model judges each topic three ways rather than one: whether the student can recall the material, work through it step by step, and apply it hands-on to a task of their own. Each is tracked separately as the student works, and a topic counts as learned only when all three are satisfied. We show that no topic can be marked as learned without evidence of every kind, whatever values the model is given. We evaluate SAGE in an exemplar programming course scenario, using simulated students, a custom educational benchmark, in-person usability sessions, and a six-week trial. Experimental results show that the standard approach marks 1999 of 2000 students who are strong on facts but weak in hands-on work as having learned the topic. Demanding more evidence without separating the kinds changes almost nothing, at 1998. Our rule marks none of them. Curriculum compliance also rises from 80% to 100% over the conference system. The same judgment also changes how the tutor teaches. Shown only a response, an independent reviewer could identify which kind of student it was written for in 83.3% of cases.**

***Index Terms*—Human-AI Teaming, Intelligent Tutoring, Knowledge Graphs, Cognitive Scaffolding, Access Control**

I. Introduction

The rapid proliferation of LLMs has transformed the landscape of Intelligent Tutoring Systems (ITS). Unlike earlier rule-based or scripted tutors, public LLMs such as ChatGPT can explain concepts, debug code, and synthesize examples in real time, creating the promise of scalable, personalized instruction for millions of learners [1]. At the same time, their widespread use in academic settings has raised unprecedented concerns about academic integrity, over-reliance on automation, and the hindrance to genuine learning habits/processes [1] [2].

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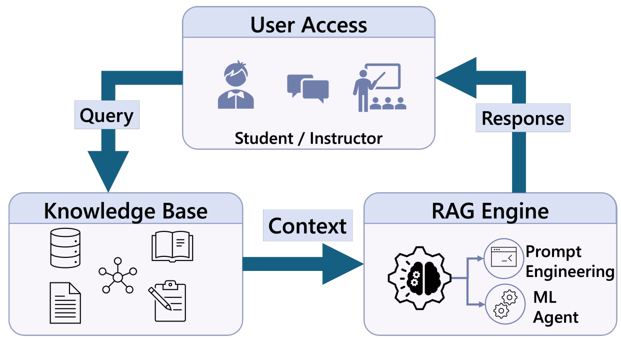

Fig. 1: Illustration of tutoring system that uses a conventional RAG approach.

At the heart of this tension is a misalignment between the objectives of general-purpose LLMs and the goals of education, particularly in STEM area course curriculum. LLMs are optimized to be helpful, i.e., to produce fluent, direct and complete answers to user prompts. Effective instruction, however, seeks to cultivate initially Instructor-Regulated Learning (IRL), where instructors actively plan, monitor and assess student learning. However, ultimately instruction could serve students' Self-Regulated Learning (SRL), in which learners actively plan, monitor, and reflect on their own learning. Instruction is ultimately in service of the learner's own regulation of their learning [3]. When an LLM returns an end-to-end solution to a task such as "How do I reverse an array?", it forecloses that development, since the learner consumes an answer rather than engaging with the reasoning that produces it, and may in any case lack the background the answer presumes.

In addition, current AI-Tutoring systems connect LLMs to external databases through Retrieval-Augmented Generation (RAG) [4] to improve accuracy. However, in most existing systems this integration is treated purely as a search task. As illustrated in **Figure 1**, a conventional "open" RAG pipeline retrieves the most semantically similar content regardless of whether it corresponds to a practice problem, an instructor solution key, or an advanced topic the student has not yet mastered. The retrieved material is then directly injected into the LLMs context [4]. From an educational perspective, this design disregards curriculum structure and learners' readiness, enabling students to bypass the productive mental struggle necessary for learning [2]. From a Human Machine Systems i.e., ITS perspective, this unrestricted access introduces a security vulnerability: sensitive instructional materials, viz. answer keys or exam materials can be exposed through ordinary dialogue [5]. Thus, what appears as a retrieval design choice simultaneously becomes a pedagogical failure and a instruction policy breach.

Crucially, prior research does not delineate learning goals and security measures as separate issues. Educational practice emphasizes *soft alignment* through prompts that ask the model to "act as a tutor" [1] [2], while security mechanisms emphasize *hard restrictions* that block access to protected data [5]. This separation leads to undesired consequences: prompts can be ignored or manipulated by students seeking answers, while rigid blocking mechanisms cannot distinguish between legitimate scaffolding and cheating. We argue that in AI-assisted pedagogical systems, there is a gap in the state-of-the-art in terms of understanding how to configure access control rules by having curriculum structure being a critical part of the system's security posture.

Our prior system, SAGE [6], addressed the instructor's side of this problem. Rather than relying on an LLM to refuse requests through prompt engineering, SAGE enforces instructional policy on a standard RAG pipeline through a three-layer Safety Gatekeeper placed between the learner and the knowledge base: (i) Role-Based Access Control (RBAC), which prevents access to unauthorized files; (ii) Attribute-Based Access Control (ABAC), which aligns content access with curriculum stage and learner progress; and (iii) Cognitive Access Control (CAC), which infers learner intent and constrains the response type, favoring hints and scaffolding over full solutions. Composed over a knowledge graph of learner state and course content, this gatekeeper reached 90% security and 80% curriculum compliance while remaining competitive in helpfulness against prompt-based tutors [6].

SAGE thus provides dependable access control over a set of topics the learner is treated as having mastered. It supplies no principled mechanism, however, for deciding what belongs in that set: membership is conferred by course completion or instructor status rather than earned through evidence of learning. The attribute on which every access decision turns the learner's mastered set is therefore asserted rather than verified. The present article supplies exactly that missing mechanism and feeds its output directly into SAGE's existing gate.

Deciding what a learner has genuinely mastered is harder than it first appears, because mastery of a programming concept is many-sided: recalling what a pointer is, tracing by hand how one behaves in a loop, and writing correct code that uses it are distinct competencies, and a learner may hold one without the others. A tutor that certifies mastery from a single kind of evidence multiple-choice recall alone, say over-credits learners who have never demonstrated the rest. We call this the *Evidence Diversity Problem*: the systematic over-crediting that arises when heterogeneous evidence is collapsed into one mastery estimate. It is a concrete instance of construct under-representation, and it is especially damaging under instructor-regulated learning, where an inflated mastery record silently corrupts the prerequisite relations the course enforces admitting a learner to a downstream topic on the strength of competence they were never shown to possess.

In this article, we extend SAGE with a learner model that turns a stream of mixed interactions into a competency-certified record of mastered concepts. For each concept the model follows three cognitive levels rather than one, routes each observation to the estimate for its own evidence type, updates those estimates online as the learner works, and certifies a concept as mastered only when all three levels are satisfied at once. Concretely, this article makes the following contributions:

- We name and define the *Evidence Diversity Problem* the over-crediting that arises when a tutor judges mastery from a single kind of evidence and show why it is especially harmful when an instructor governs progression, since an inflated mastery record corrupts the prerequisites the course depends on.
- We introduce a *tier-factored knowledge-tracing learner model* that routes each observation to the estimate for its evidence type, updates online, and grants mastery only under a conjunctive rule over all three levels, producing a mastered set that an instructor-regulated gate can consume directly.
- We prove an *Evidence Diversity Guarantee*: a concept cannot be certified unless the system has observed at least $n_{\min}$ correct observations of *every* required evidence type a property of the architecture itself, independent of the values assigned to its parameters and we complement it with quantitative bounds on false certification.
- We integrate this model into SAGE so that its prerequisite gate acts on *earned rather than asserted* mastery, and add a mastery-conditioned response-adaptation policy that fades scaffolding as mastery grows, together with an instructor risk dashboard that surfaces at-risk learners for intervention.

Our evaluation proceeds in three parts. We first isolate the learner model in a controlled simulation harness, measuring how often a concept would have been certified from a single evidence type and how the certification rule behaves as its parameters are varied. We then evaluate the system with human participants in individual, task-based usability sessions [PLACEHOLDER: under IRB approval, $N = \ldots$ learners and $N = \ldots$ instructors], and report on the perceived usefulness of the adaptive tutor and the actionability of the instructor dashboard. Finally, we deploy the system for a six-week beta period [PLACEHOLDER: $N = 10$ participants] in which learners use it for their own coursework. This last setting allows us to observe retention and decertification over the intervals on which forgetting actually operates, and to measure over-refusal on legitimate curriculum vocabulary

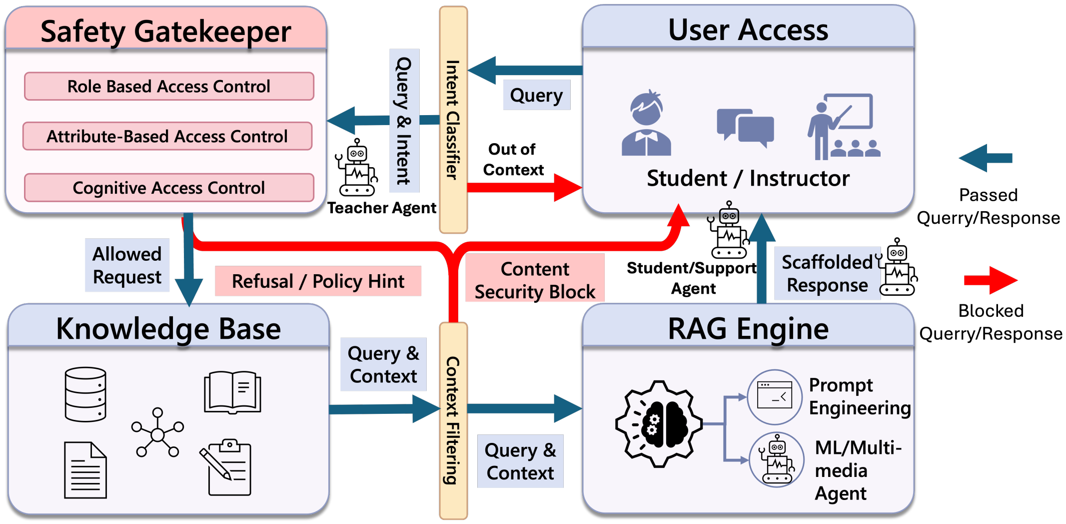

Fig. 2: Overview of SAGE for intelligent tutoring that shows the interaction flow and components for meeting learning goals and enforcing security.

in authentic use. Throughout, we retain the benchmark comparison against prompt-based LLM baselines and report an expanded adversarial audit of the inherited gate, which we have since hardened against the multi-turn attack surface that per-message enforcement leaves open.

The remainder of this article is organized as follows. Section II reviews related work on instructor-regulated tutoring, knowledge tracing, and cognitive diagnosis. Section III recaps the inherited SAGE architecture, formalizes the factored learner model, the certification rule, and the Evidence Diversity Guarantee, and describes the adaptation policy, supporting components, and instructor dashboard. Section IV presents the evaluation across simulation, usability sessions, and beta deployment, and Section V concludes.

II. Related Work

A. Intelligent Tutoring Systems

Regulated learning is a critical component for academic success, traditionally viewed through Zimmerman's cyclical model [3]. It consists of learnable processes, such as setting goals and self-monitoring, that are heavily influenced by instructional scaffolding and feedback [7] [8]. Early ITS utilized rule-based pedagogical policies to explicitly prompt learning behaviors such as help-seeking and metacognitive monitoring. While these scaffolds effectively improve learning outcomes and regulatory skills [9], [10], their reliance on manual authoring and rigid domain modeling limits scalability. Conversely, LLMs offer scalable, personalized instruction through natural-language interaction [1]. However, the shift toward generative AI-based tutors necessitates new mechanisms to ensure these systems actively support, rather than bypass, a learner's internal regulatory processes.

Further, recent studies caution that general-purpose LLMs are not inherently aligned with pedagogical objectives as they prioritize providing complete answers over instructional value [1]. This tendency can allow learners to bypass the productive struggle and metacognitive engagement essential for instructor- or self-regulated learning. Empirical evidence also confirms that the pedagogical impact of AI chatbots depends heavily on interaction design. Towards this, a meta-analysis indicates that learning outcomes improve when interactions provide guided support; conversely, unguided answer-giving can negatively affect higher-order problem-solving [2]. Moreover, systematic reviews suggest that while LLMs enhance accessibility, they risk fostering a reliance on the model that diminishes a learner's internal regulatory processes when deployed without explicit instructional constraints [11]. We address such concerns in our work through investigations on how interaction structures can be designed to scaffold planning, monitoring, and reflection behaviors, therefore supporting learners' ability to learn when engaging with general-purpose LLM-based tutors.

B. Retrieval-Augmented Generation in Educational Tutoring

To mitigate hallucinations and ground responses in course materials, many tutoring systems utilize RAG [4].

While RAG improves factual grounding in learning-oriented settings [12], semantic similarity is often pedagogically agnostic. It fails to account for prerequisite structures or curriculum sequencing; consequently, the most "relevant" retrieved document may be an answer key or advanced material that bypasses the necessary learner's struggle. Beyond pedagogical concerns, the retrieval layer in RAG presents a significant security risk as well. Recent research [5] indicates that retrieved context is a primary leakage channel where prompt injections can manipulate LLMs into revealing sensitive or restricted information once it enters the context window. Since generation-layer safeguards are often probabilistic and bypassable, these risks motivate the enforcement of constraints at the retrieval boundary rather than relying solely on language-level refusals.

Moreover, current tutoring approaches typically treat learning goals and security mechanisms as separate problems. While educational research investigates the potential of LLMs to enhance regulated learning and metacognitive support [1], [2], the security community emphasizes the risks associated with prompt injection and data poisoning in these integrated systems [5]. The disconnect between these domains creates a significant limitation in existing platforms, as the absence of integrated access controls prevents the enforcement of structured, step-by-step guidance when students access RAG implementations for course content. Existing solutions for content moderation often rely on Reinforcement Learning from Human Feedback [13] or explicit regex-based content filters at the generation layer. However, these safeguards do not change underlying LLMs' access to information; therefore, as long as sensitive content is present in the retrieval context, there remains a non-zero probability that it will be exfiltrated under adversarial prompting [5].

In addition to the advances in RAG, recent work on GraphRAG that uses knowledge graphs has demonstrated improved controllability by encoding concept dependencies and curriculum structure that flat vector similarity ignores [14]. There has been further evidence that shows using knowledge graphs can enhance interpretability and alignment in sequenced learning domains [15]. However, existing GraphRAG and secure RAG frameworks typically treat retrieval as a neutral preprocessing step, rather than an explicit component of the instructional design. From a security perspective, this limitation has motivated a shift toward restricting LLM access to data, rather than relying solely on behavioral alignment, to mitigate risks such as prompt injection [5]. The above works motivate our Safety Gatekeeper-based approach in which retrieval itself is governed by pedagogical and organizational policy, enabling access to be conditioned on instructional intent and curriculum constraints.

Enforcement at the retrieval boundary is conventionally applied one message at a time. A per-message decision, however, cannot detect an attack that is assembled across a conversation. The crescendo threat model [16] demonstrates this directly: an adversary decomposes a prohibited goal into individually innocuous turns, and the harmful payload is assembled only by a final instruction that carries no indication of intent on its own. Recent work on agent memory reaches a similar conclusion in a different setting, showing that auditing records in isolation cannot detect content whose harm emerges only in a particular context, and therefore validating each record against the surrounding interaction history [17]. Towards this, session-level risk aggregation has been proposed as a training-free defense that escalates suspicious conversations to a conversation-level adjudicator [18]. We adopt this posture in Section III-C, where the unit of enforcement is the session rather than the query.

C. Learner Modeling: Knowledge Tracing and Cognitive Diagnosis

Estimating what a student knows from a sequence of responses is the province of *knowledge tracing*. Bayesian Knowledge Tracing (BKT) [19] models each skill as a two-state hidden Markov process and updates a posterior probability of mastery after every response, giving a transparent, online estimate that has anchored mastery-based progression in intelligent tutors for three decades. Deep Knowledge Tracing (DKT) [20] and its successors replace the per-skill Markov assumption with a recurrent network, improving predictive accuracy by sharing statistical strength across skills at the cost of interpretability and a substantial data requirement. For all their differences, both families share a property that matters here: they summarize a learner's competence on a skill as a *single* scalar, collapsing recall, procedural fluency, and applied problem-solving into one number. A learner who answers multiple-choice questions well but cannot yet write working code is indistinguishable, to such a model, from one who can do both precisely the over-crediting the Evidence Diversity Problem describes, and a textbook case of construct under-representation in measurement terms [21].

Cognitive diagnosis models (CDMs) take the complementary step of separating competence into multiple latent attributes. The DINA model [22], [23], in particular, applies a *conjunctive* rule: a task is answered correctly only if the learner possesses *all* of its required attributes, so mastery of a compound skill demands every component at once. This is the same all-or-nothing intuition our certification rule rests on, and we claim no novelty for it. CDMs, however, are designed for after-the-fact psychometric diagnosis: their attributes are estimated in batch from a fixed response matrix calibrated against a hand-authored Q-matrix, they do not update online as a student works, and critically for our setting they do not route evidence by *type*, treating all items that load on an attribute as interchangeable. They tell an analyst which attributes a cohort lacks; they do not hand a tutor a running, evidence-diverse mastery signal on which to gate the next interaction.

Conjunctive mastery criteria have also been applied within knowledge tracing itself. Conjunctive knowledge tracing [24] requires all skills attached to an item to be mastered before the item is retired, and multi-skill and dynamic-Bayesian variants of BKT jointly model several knowledge components rather than one per model [25]. Closest to our motivation, Huang et al. [26] observe that tracing decomposed component skills alone risks premature assertion of mastery, and grant mastery only when a learner demonstrates a skill in combination with others, evaluated on a Java programming tutor. We claim no novelty for conjunctive mastery as such. What distinguishes the present model is the axis along which evidence is factored: prior work factors by knowledge component, several distinct skills, whereas we factor a *single* concept by mode of demonstration, so

TABLE I: Capabilities of representative learner-modeling and tutoring approaches. ✓: supported; △: supported only by some variants; ×: not supported.

Capability	BKT	DKT	CDM	SAGE	Ours
Online updating	✓	✓	△	×	✓
Multi-dimensional mastery	×	×	✓	×	✓
Evidence-type routing	×	×	×	×	✓
Conjunctive mastery	×	×	✓	×	✓
Curriculum enforcement	×	×	×	✓	✓
Mastery-conditioned adaptation	×	×	×	×	✓

that recall, procedural fluency, and applied competence on the same topic are tracked as separate evidence types and required jointly. The resulting certified set is then consumed directly by a retrieval-time access-control decision, which none of the preceding models supply.

D. Mastery-Based Adaptation and Expertise Reversal

Once a mastery estimate exists, a tutor must decide how to act on it. The *expertise-reversal effect* [27] establishes that instructional support which helps novices worked examples, heavy scaffolding becomes redundant or even counterproductive as competence grows, motivating adaptation that fades support in step with mastery and keeps instruction within the learner's zone of proximal development [28]. Our mastery-conditioned response-adaptation policy (Section III) applies this principle to an LLM tutor, mapping the certified mastery level to scaffolding depth, response format, and challenge type. We treat this as a faithful implementation of an established principle rather than a new mechanism; its role here is as the consumer that turns the certified signal into pedagogical action.

E. Summary of the Gap

Table I places these strands side by side. The two capabilities the Evidence Diversity Problem demands are each present in prior work, but never together. Knowledge tracing updates a mastery estimate online as a student works, but collapses every kind of evidence into one score. Cognitive diagnosis models separate competence into several dimensions and can require all of them at once, but are built for after-the-fact diagnosis rather than tutoring control, and do not route evidence by type. And instructor-regulated tutors such as SAGE can act on a mastered set but cannot themselves determine what belongs in it. The learner model of this article brings these strands together routing each kind of evidence to its own estimate, updating those estimates online, and granting mastery only when all are met so that the resulting record of mastered topics can be trusted by an instructor-regulated gate.

III. SAGE Methodology

A fundamental challenge in the deployment of general-purpose LLMs for education is the *alignment gap* between a Large Language Model's stochastic helpfulness objective and the rigid constraints of an academic curriculum. SAGE addresses this by transforming the LLM tutor into a *stateful component* of an ITS, governed by a deterministic Safety Gatekeeper as shown in **Figure 2** that controls what information is available to the LLM's context. This design enables the SAGE to respond naturally to user queries while i) preserving the learner's regulated learning cycle, and ii) strictly enforcing institutional policies regarding assessment content and curriculum pacing. We model the learner session state as:

$$S = \{K, R, C\} \tag{1}$$

where K denotes the set of mastered knowledge nodes (topics), R is the user role (e.g., student or instructor), and C denotes the current cognitive intent (e.g., scaffolding request, solution seeking, or administrative command). Each incoming query q updates S and determines whether access to a candidate document d is permitted. In ITS terms as a human–machine system, the LLM operates as an *actuator* within a closed control loop, where learner actions shape the session state S, thereby constraining what the LLM can retrieve and generate.

A. Dual-Store Knowledge Architecture

SAGE employs a hybrid knowledge architecture that integrates unstructured semantic retrieval with structured curriculum modeling. Unlike flat vector retrieval, a GraphRAG design is used, consisting of two coupled stores: i) A **Vector Database** V (ChromaDB), which indexes chunked instructional content as dense embeddings for semantic similarity retrieval and ii) A **Knowledge Graph** $G = (T, E)$ (Neo4j), where T is the set of topic nodes and E encodes prerequisite relationships in a Directed Acyclic Graph (DAG). Each topic $t \in T$ stores difficulty, instructor-defined learning status, and mastery metadata.

Given a query q, initial candidate chunks are retrieved via:

$$\mathcal{D}_{cand} = \text{topK}(V, \text{embed}(q)). \tag{2}$$

Named entities and topic mentions are mapped to nodes in G. For a topic t, prerequisites are:

$$\text{Prereq}(t) = \{t' \in T \mid (t', t) \in E\}. \tag{3}$$

Given learner knowledge state K, readiness is defined as:

$$\text{Ready}(t, K) = \begin{cases} 1, & \text{if } \text{Prereq}(t) \subseteq K, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Thus retrieval becomes *state-aware*: if q requests "arrays" but the learner lacks proficiency in "loops," then $\text{Ready}(\text{Arrays}, K) = 0$ and the system redirects toward prerequisite scaffolding rather than providing a full solution. *Human-in-the-Loop Control:* Instructors can modify topic attributes and prerequisite relations within the Knowledge Graph serving as a "human-in-the-loop control", and these modifications are immediately reflected in the Safety Gatekeeper decisions during subsequent queries as shown in **Figure 3**. An instructor-facing interface is used to allow

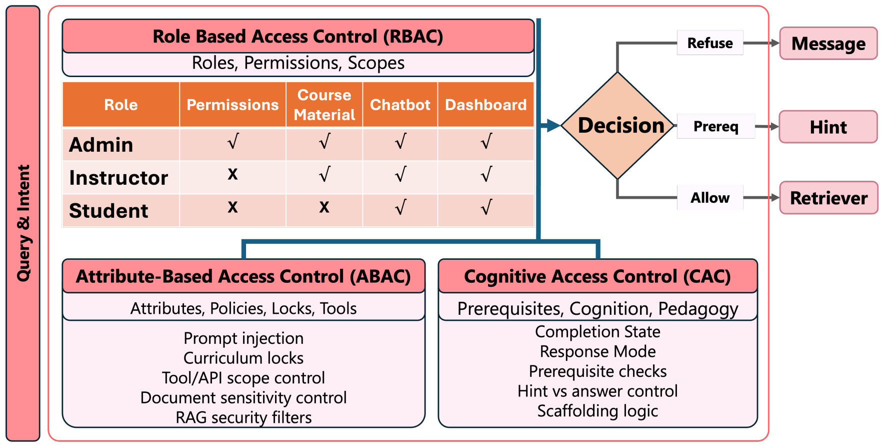

Fig. 3: SAGE security architecture to enforce how the system routes user queries through a Safety Gatekeeper that incorporates semantic analysis and specific access control protocols, including RBAC, ABAC, and CAC.

authorized users (i.e., instructors) to lock or unlock topics, adjust prerequisite relationships, and flag sensitive materials within the Knowledge Graph. These updates propagate immediately to the Safety Gatekeeper, enabling real-time policy enforcement without redeploying LLMs or retraining classifiers. This design supports dynamic curriculum management and allows instructors to intervene when pedagogical goals or assessment policies change.

Thus, SAGE operates over an instructor-controlled course corpus containing lecture notes, practice materials, and restricted assessment artifacts. The corpus is chunked into passages for vector indexing, and each chunk d is annotated with: (i) a type label $type(d)$ such as lecture, practice, solution, exam key for RBAC; (ii) an associated curriculum topic t when applicable for ABAC; and (iii) a response mode $mode(d)$ such as conceptual explanation, hint, partial solution, and instructor-only for CAC. These annotations are produced via an instructor-facing interface and validated by spot checks.

Cross-Turn Topic Resolution: State-aware retrieval as formulated in Eq. (4) assumes that each query names the concept it concerns. Real tutoring dialogue, however, is anaphoric. Following a definition of arrays, a learner asks "how does it work?", "how do I declare it?", or "what about a nested one?", none of which names a topic. Left unresolved, such a turn either retrieves the wrong concept or reaches the gatekeeper as a context-free fragment and is misread as off-topic. We therefore resolve reference deterministically rather than by inference. The session retains an anchor t_{anchor}, seeded from the learner's own concept at the earliest reliable point in the session, and a vague or anaphoric follow-up is rewritten against this anchor before any access decision is taken. The rewrite is guarded so that it does not fire on an explicit topic switch or on an emotional message, and it runs before the gatekeeper so that $\mathrm{Ready}(t, K)$ is evaluated on the intended concept. This is standard dialogue-state maintenance and we claim no novelty for it; we report it because state-aware retrieval is not realizable on real dialogue without it.

B. Safety Gatekeeper and Access Control

SAGE security architecture in the three-layer Safety Gatekeeper is shown in **Figure 3**. Herein, we discuss access control for each candidates query, where the retrieved data chunks are represent by $d \in \mathcal{D}_{cand}$:
Role-Based Access Control (RBAC): Each of the retrieved chunks for the query carries a type label by design such as lecture note, practice solution, exam key, etc. The access to these chunks are permitted if:

$$\mathrm{RBAC}(d, R) = \begin{cases} 1, & \text{if role\_perm}(R, \text{type}(d)) = \text{allow}, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

For instance, exam answer keys are always denied for role $R = student$, thus removing sensitive content from possible exposure.
Attribute-Based Access Control (ABAC): Subsequently, if the retrieved chunks d corresponds to topic t, access to them depends on curriculum state and mastery i.e., for example whether the topic is open or not.

$$\mathrm{ABAC}(d, K) = \begin{cases} 1, & \text{if status}(t) = \text{Open} \wedge \text{Ready}(t, K) = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

The instructors may lock topics or enforce pacing constraints, ensuring zone-of-proximal-development alignment.
Cognitive Access Control (CAC): We map four anchors into a conservative access decision by checking whether the retrieved document's *response mode* is compatible with the inferred intent. Towards this, a lightweight intent model infers a coarse tutoring intent label $C \in \{conceptual, problem\_solving, debugging, review\}$:

$$CAC(C, d) = \begin{cases} 1, & \text{if } mode(d) \in \mathcal{M}(C) \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

where $\mathcal{M}(C)$ encodes organizations' tutoring policy for example, *problem_solving* permits hints or partial code, while instructor-only solution keys remain disallowed for students. This formulation subsumes the binary distinction between *solution-seeking* and *scaffolding-oriented* requests by treating *problem_solving* as the solution-adjacent anchor and the remaining intents as scaffolding-oriented.

Implementation Details: The CAC layer performs low-latency intent classification using the CPU-based sentence embedding model `all-MiniLM-L6-v2` with 384-dimensional embeddings. The intent is inferred via cosine-similarity matching against coarse anchors for conceptual inquiry, problem solving, debugging, and review, with conservative heuristic overrides for security-sensitive queries. Domain-specific entity extraction is handled separately using `gliner_small-v2.1`, a zero-shot span classification model that supports custom labels without retraining. This enables robust identification of programming constructs and error types that are poorly handled by conventional NER methods. CAC decisions are intentionally conservative and composed conjunctively with RBAC and ABAC constraints.

As a result, misclassification can affect the instructional responses, for example level of scaffolding or explanation depth, but does not affect content access or retrieval, which is primarily governed by RBAC and ABAC.

Context-Gated Classification: The conservative overrides that handle security-sensitive queries are, by contrast, capable of denying a response outright, and a blunt formulation of them over-refuses core curriculum vocabulary. Terms such as "infinite loop" and `fork()` denote concepts an introductory C learner must study in order to recognize and avoid, yet a phrase match alone blocks "why does my code have an infinite loop?", a debugging request, as a security risk. Crucially, a false block in a tutoring system carries a direct pedagogical cost that a generic deployment does not incur: the learner is refused instruction on material the course itself requires them to master. We therefore gate these overrides on co-occurrence rather than on presence. A curriculum term is treated as adversarial only when it appears together with a harmful goal such as *crash*, *freeze*, or *never stop* in the same message, while unambiguous attack phrases such as *fork bomb* are blocked regardless of context. For the same reason, the overrides are keyed on explicit rules rather than on embedding similarity, since a fuzzy security anchor false-matches innocuous tokens and reintroduces the false blocks the gating is intended to remove. Allowing a benign single turn to proceed is what permits the session-level layer of Section III-C to adjudicate the conversation in which it sits.

As a result of the three-layered Safety Gatekeeper, a document d is admitted to the LLM context only if

$$\text{Allow}(d \mid S) = \text{RBAC}(d, R) \wedge \text{ABAC}(d, K) \wedge \text{CAC}(C, d) \quad (8)$$

A document is therefore included in the context window iff $\text{Allow}(d \mid S) = 1$. **Algorithm 1** summarizes this query-handling loop, showing how RBAC, ABAC, and CAC are composed to identify the appropriate context window. Once the context is defined, the system delegates control to different pedagogical agents that control information presentation to the learner and provide regulatory control to instructors.

Algorithm 1 SAGE Query Processing

Require: Query q, session state $S = \{K, R, C\}$
Ensure: Tutor response y
// Semantic analysis
$C \leftarrow \text{IntentClassifier}(q)$
$\mathcal{T}_q \leftarrow \text{ExtractTopics}(q)$ ▷ map to KG topics
$\mathcal{D}_{cand} \leftarrow \text{topK}(V, \text{embed}(q))$ ▷ vector retrieval
// Prerequisite & curriculum checks
for all $t \in \mathcal{T}_q$ **do**
    $\text{ready}[t] \leftarrow \text{Ready}(t, K)$ ▷ Eq. (4)
end for
// Safety Gatekeeper filtering
$\mathcal{D}_{allow} \leftarrow \emptyset$
for all $d \in \mathcal{D}_{cand}$ **do**
    $r \leftarrow \text{RBAC}(d, R)$
    $a \leftarrow \text{ABAC}(d, K)$
    $c \leftarrow \text{CAC}(C, d)$
    if $r \wedge a \wedge c = 1$ **then**
        $\mathcal{D}_{allow} \leftarrow \mathcal{D}_{allow} \cup \{d\}$
    end if
end for
// No safe context available
if $\mathcal{D}_{allow} = \emptyset$ **then**
    if $\exists t \in \mathcal{T}_q : \text{ready}[t] = 0$ **then**
        return PREREQUISITEHINT$(\mathcal{T}_q, K)$
    else
        return REFUSALMESSAGE(q)
    end if
end if
// Agent selection and generation
if $R = \text{student}$ **then**
    $y \leftarrow \text{SocraticStudentAgent}(q, \mathcal{D}_{allow}, S)$
else
    $y \leftarrow \text{TeacherAdminAgent}(q, \mathcal{D}_{allow}, S)$
end if
return y

C. Trajectory-Level Access Control

Each of RBAC, ABAC, and CAC in Eq. (8) evaluates a single candidate document against a single query. This posture is blind to crescendo attacks [16], in which a prohibited objective is decomposed across turns and the assembling instruction is superficially benign. Herein, we adapt the peak-and-accumulation approach of [18] to the access-control boundary of an instructor-regulated tutor. We retain its central idea, viz. that conversation-level risk should combine the worst single turn with the persistence of suspicion across turns, while the aggregation below differs from theirs and is instantiated on signals the tutoring pipeline already computes. The per-message gate is retained unchanged, and a session-level risk state is layered over it to determine when a conversation-level adjudication is performed.

At turn t, a scalar risk $r_t \in [0, 1]$ is computed from signals the pipeline already produces: i) the goal-alignment gap $(1 - s_{\text{goal}})$, ii) drift from the course domain, iii) accumulated off-topic strikes, and iv) proximity to attack vocabulary. No additional inference is introduced. The session retains two scalars, updated as:

$$\mathrm{peak}_t \leftarrow \max\big(\lambda\,\mathrm{peak}_{t-1},\ r_t\big)\,, \tag{9}$$

$$\mathrm{acc}_t \leftarrow \min(1,\ \lambda\,\mathrm{acc}_{t-1} + r_t)\,, \tag{10}$$

with $\lambda_{\mathrm{peak}} = 0.9$ and $\lambda_{\mathrm{acc}} = 0.8$, the peak cooling more slowly so that a single sharp excursion remains visible for several subsequent turns, and the session risk is their equal combination:

$$R_t = \tfrac{1}{2}\,\mathrm{peak}_t + \tfrac{1}{2}\,\mathrm{acc}_t. \tag{11}$$

The two terms capture complementary attack shapes, i.e., a single sharp excursion and a slow accumulation of individually unremarkable turns. We weight them equally because neither is a priori more concerning. While $R_t < \tau_{\mathrm{judge}}$, the session proceeds under the per-message gate alone. Once $R_t \geq \tau_{\mathrm{judge}}$, the turn is escalated to a conversation-level judge that reads the last six turns of the transcript, so that a benign final instruction is adjudicated in the context of the conversation that produced it.

Block Poisoning: A subtlety arises from the ordering of the two layers. A per-message block returns before the session state is updated, so a blocked malicious turn would otherwise leave the accumulator clean and allow an immediate reformulation to pass unexamined. Every security-grade block therefore raises the accumulator floor, and subsequent turns in the same session are scrutinized. A learner who returns to legitimate work is still permitted to continue, since the judge rather than a fixed rule arbitrates the escalation.

Trigger versus Classifier: The accumulator is a trigger and not a classifier, and we frame it accordingly. The per-turn risk does not separate adversarial from legitimate sessions, since a debugging session saturated with loop-termination language and a legitimate systems-programming session both accumulate risk within the range attained by attacks. The design does not require separation. The accumulator determines when adjudication is worth its cost, and the transcript-fed judge performs the classification. Thus τ_{judge} trades judge invocations against the opportunity to adjudicate, and is not a correctness boundary. We report the aggregate behavior of the extended gate in Section IV-D, and note there that the residual attack surface consists of operations expressed in unmarked curriculum vocabulary, which the present per-turn scorer does not detect.

The layer is inexpensive by construction. Its entire state is two scalars per session; it introduces no model call on the common path, and the judge is consulted only once accumulated risk is elevated. This preserves the negligible security overhead reported for the inherited gate.

D. Multi-Agent Pedagogical Interaction

We use the term *multi-agent* to denote a set of role-specialized pedagogical agents (student, teacher/admin, and multimedia support) orchestrated by the Safety Gatekeeper, rather than fully autonomous agents with independent goals or learned policies. The system comprises three role-aligned agents:

1) *Socratic Student Agent*: generates structured hints, logic plans, and optionally flow diagrams rather than complete solutions, so that the learner performs the reasoning the task requires.
2) *Administrative Teacher Agent*: operates with elevated privileges, enabling curriculum manipulation for example locking/unlocking modules and access to restricted assets via secure function calls.
3) *Multimedia Support Agent*: generates visual scaffolds such as state diagrams, memory models, control-flow charts to assist conceptual reasoning and neurodiverse learners.

Because agent selection is itself governed by the session state S, the Safety Gatekeeper acts as a learning-aware controller rather than a generic content filter: which agent responds, and with what latitude, follows from the learner's role and certified mastery rather than from the surface form of the query. All agents work in synchronization with the Safety Gatekeeper and dual-store knowledge architecture, ensuring consistency of instructor-defined curriculum constraints across learning sessions.

E. The Learner Model: Factored Knowledge Tracing

Thus far, the mastered set K has been treated as given. In SAGE as originally formulated, membership in K is conferred by course completion or instructor status: the gate is sound, but the attribute it acts on is asserted. We now specify the learner model that *earns* K from a stream of mixed interactions. Its output is the same object the gate already consumes the set K of Eq. (1), read by $\mathrm{Ready}(t, K)$ in Eq. (4) and by $\mathrm{ABAC}(d, K)$ in Eq. (6) so the model changes the *provenance* of K without altering a single gate equation. We design the model to three ends: it must update online as the learner works, it must distinguish kinds of evidence rather than pooling them, and it must certify mastery under a rule an instructor-regulated gate can trust.

Mastery of a programming concept is not monolithic: recalling a definition, writing a single correct line, and debugging working logic are distinct competencies. We therefore track each topic $t \in T$ not with one estimate but with three, one per cognitive tier $k \in \{1, 2, 3\}$ corresponding to declarative, procedural, and applied understanding. Each tier maintains an unconstrained Bayesian Knowledge Tracing posterior $\tilde{P}_t^{(k)} \in [0, 1]$, the probability that the learner has mastered topic t at tier k.

Evidence Routing: Each observation the learner produces is an assessed interaction $o = (t, k, e)$, where t is the topic, $e \in \{0, 1\}$ the graded outcome, and $k = \rho(o)$ the tier assigned by a deterministic routing function ρ. The routing is by instrument rather than by inference: a multiple-choice or short-answer item routes to the declarative tier, a single-line completion or trace to the procedural tier, and a submitted

TABLE II: Per-tier learner-model parameters. The orderings are structural rather than fitted: guess rates fall across the tiers, since recognition items admit more guessing than working code, while slip rates rise, since an applied task affords more ways to err despite knowledge. Priors fall likewise, since incoming learners plausibly carry partial declarative exposure but near-zero applied skill. The transition rate is uniform across tiers. Ceilings weight the composite display estimate of Eq. (15), sum to 0.95 by construction, and are excluded from certification.

Parameter	Declarative	Procedural	Applied
Guess $p_G^{(k)}$	0.20	0.10	0.05
Slip $p_S^{(k)}$	0.10	0.15	0.20
Prior $p_{L_0}^{(k)}$	0.30	0.05	0.01
Transition $p_T^{(k)}$	0.09	0.09	0.09
Decay half-life (days)	14	7	5
Ceiling $\theta_{\max}^{(k)}$	0.60	0.25	0.10

program or debugging task to the applied tier. Crucially, ρ is total and single-valued, so every observation updates exactly one tier and no observation updates two. This is the property on which the guarantee of Section III-F1 rests.

Per-Tier Update: Let $P_t^{(k)}$ denote the posterior for topic t at tier k before an observation, and let the tier carry parameters $(p_G^{(k)}, p_S^{(k)}, p_{L_0}^{(k)}, p_T^{(k)})$ for guess, slip, prior, and transition. On an observation routed to tier k, the posterior is first conditioned on the outcome:

$$P_{t|e=1}^{(k)} = \frac{P_t^{(k)}\,(1-p_S^{(k)})}{P_t^{(k)}\,(1-p_S^{(k)}) + (1-P_t^{(k)})\,p_G^{(k)}}, \tag{12}$$

$$P_{t|e=0}^{(k)} = \frac{P_t^{(k)}\,p_S^{(k)}}{P_t^{(k)}\,p_S^{(k)} + (1-P_t^{(k)})\,(1-p_G^{(k)})}, \tag{13}$$

and then advanced by the learning transition, which is applied on correct responses only:

$$\tilde{P}_t^{(k)} \leftarrow \begin{cases} P_{t|e}^{(k)} + \left(1-P_{t|e}^{(k)}\right) p_T^{(k)}, & e=1, \\ P_{t|e}^{(k)}, & e=0. \end{cases} \tag{14}$$

Crediting learning after an incorrect attempt would permit wrong answers to drift the posterior upward, so the transition is withheld in that case.

The emission parameters are set per tier rather than shared, since the tiers differ in how they can be answered without competence. Guessing is easier on a recognition item than in working code, so guess rates fall across the tiers, while an applied task affords more ways to err despite knowledge, so slip rates rise. The transition rate is uniform across tiers, since we have no principled basis for supposing that learning proceeds faster at one level of understanding than another. Table II gives the deployed values.

Composite Display Estimate: For presentation to the learner, the three posteriors are summarized as a single composite,

$$\bar{P}_t = \sum_k \theta_{\max}^{(k)}\,\tilde{P}_t^{(k)}, \qquad \textstyle\sum_k \theta_{\max}^{(k)} = 0.95, \tag{15}$$

where the per-tier ceilings $\theta_{\max}^{(k)}$ of Table II act as weights, bounding each tier's contribution to the composite so that no single evidence type dominates the displayed estimate. Note that the declarative tier is weighted most heavily, which is appropriate for a progress indicator a learner consults early and often, but is precisely the weighting the Evidence Diversity Problem warns against for a mastery judgment. Crucially, $\bar{P}_t$ is a presentation quantity only. Certification (Section III-F1) is evaluated on the raw posteriors $\tilde{P}_t^{(k)}$, never on the composite, so no choice of $\theta_{\max}^{(k)}$ can admit a topic to K.

F. Conjunctive Certification and the Earned Mastered Set

A topic is certified as mastered only when *every* tier is satisfied at once. Formally, $t \in K$ iff

$$\underbrace{\forall k:\ \tilde{P}_t^{(k)} \geq \theta_{\text{cert}}}_{\text{all tiers confident}} \quad \wedge \quad \underbrace{\forall k:\ N_t^{(k)} \geq N_{\min}}_{\text{all tiers evidenced}}, \tag{16}$$

with $N_{\min} = 3$ and a per-concept, per-tier certification bar $\theta_{\text{cert}}^{(k)} \geq 0.95$, floored at that value and raised where an instructor or a calibration procedure requires greater confidence for a particular concept. Theorem 1 quantifies over all admissible bars, so tightening cannot weaken the guarantee. Certification is evaluated on the raw posteriors; the ceilings $\theta_{\max}^{(k)}$ of Table II shape only the composite score shown to the learner, so an early-stage learner is never displayed high applied mastery even while the underlying posterior is still accumulating. To prevent oscillation at the boundary, certification is governed by Schmitt-trigger hysteresis: once certified, a topic is *de*certified only if some tier decays below $\theta_{\text{dec}} = 0.75 < \theta_{\text{cert}}$.

The certified set $K = \{t \in T : t \text{ satisfies Eq. (16)}\}$ is precisely the mastered set of the session state $S = \{K, R, C\}$ in Eq. (1). It is consumed unchanged by $\text{Ready}(t, K)$ (Eq. (4)) and $\text{ABAC}(d, K)$ (Eq. (6)); the Safety Gatekeeper is not modified. The learner model's sole effect on access control is to replace an asserted K with an earned one, so that every prerequisite and pacing decision the gate makes now rests on demonstrated, evidence-diverse competence.

1) The Evidence Diversity Guarantee

The certification rule of Eq. (16) is not merely a conservative threshold. Together with the routing of Section III-E it yields a structural property: certain evidence profiles cannot certify a topic at all, irrespective of how much evidence they contain and irrespective of the parameter values in Table II. We state this precisely.

Lemma 1 (Tier Isolation)**.** *For any observation sequence O and any tier k, the trajectory of $\tilde{P}_t^{(k)}$ and of $N_t^{(k)}$ depends only on the subsequence $O^{(k)} = \{o \in O : \rho(o) = k\}$. Observations routed to tiers $j \neq k$ leave both quantities unchanged.*

Proof. By construction of the update. An observation o with $\rho(o) = j$ applies Eqs. (12)–(14) to $\tilde{P}_t^{(j)}$ and increments $N_t^{(j)}$ alone; no term in either expression references the state of any other tier, and ρ is single-valued, so o contributes to exactly one subsequence. The decay of Eq. (18) is likewise applied per tier with per-tier rate λ_k. Hence the tier-k state is a function of $O^{(k)}$ only. □

Theorem 1 (Evidence Diversity Guarantee)**.** *Let O be any observation sequence for topic t and let $n_k = |O^{(k)}|$ be the number of observations routed to tier k. Then:*

(i) *If $n_k < N_{\min}$ for some tier k, then $t \notin K$, for every assignment of the parameters of Table II and for every $|O|$.*

(ii) *If $O^{(k)}$ contains no correct observation, then $t \notin K$, again for every assignment of the parameters of Table II and irrespective of n_k.*

Proof. For (i), Eq. (16) requires $N_t^{(k)} \geq N_{\min}$ for every k. By Lemma 1, $N_t^{(k)} = n_k$, since only observations in $O^{(k)}$ increment it. If $n_k < N_{\min}$ the second conjunct of Eq. (16) fails and certification is refused. No observation outside $O^{(k)}$ can increment $N_t^{(k)}$, so enlarging O elsewhere cannot repair the deficit, and the argument uses no property of the parameters.

For (ii), suppose $O^{(k)}$ consists entirely of incorrect observations. Under Eq. (14) no learning transition is applied, so the tier posterior evolves solely by conditioning on $e = 0$. Non-degeneracy of the emission parameters, $p_S^{(k)} + p_G^{(k)} < 1$, makes this map a strict contraction toward zero, so the sequence is monotone decreasing from $p_{L_0}^{(k)}$ and converges to 0. The decay of Eq. (18) pulls toward $p_{L_0}^{(k)} < \theta_{\text{cert}}$ and so cannot raise the posterior above any bound it does not already exceed. Hence $\tilde{P}_t^{(k)}$ never attains θ_{cert}, the first conjunct of Eq. (16) fails, and $t \notin K$. □

Corollary 1 (Minimum Diverse Evidence)**.** *Certification of a topic requires at least $N_{\min}$ observations in each of the three tiers, and therefore at least $3N_{\min}$ observations in total. Under the deployed parameterization the two conjuncts of Eq.* (16) *bind simultaneously: three consecutive correct observations carry the declarative, procedural, and applied posteriors to* 0.982*,* 0.981*, and* 0.989 *respectively, so nine correct observations, three of each kind, is the exact minimum for certification.*

Remark: Both parts hold for every admissible parameterization, which is what permits the parameter discussion of Section III-K to defend the constants by their consequences rather than by fit. The guarantee is a property of the architecture, i.e. of the routing, the conjunction, and the withheld transition, and not of any value the model was configured with. The withheld transition is essential to part (ii). Were p_T applied irrespective of outcome, an all-incorrect sequence would converge to a nonzero fixed point, 0.103, 0.108, and 0.114 for the three tiers, and part (ii) would hold only under the additional condition that these lie below θ_{cert}. Withholding the transition removes the condition, since the incorrect-response map is then a strict contraction toward zero.

G. Mastery-Conditioned Response Adaptation

Certification governs what the tutor may *retrieve*; the same mastery signal also governs how it should *respond*. The expertise-reversal effect [27] holds that the heavy scaffolding which helps a novice becomes redundant, and eventually obstructive, as competence grows. We operationalize this with a policy Π that maps the learner's mastery on the current topic to a response configuration scaffolding depth, explanation style, and challenge type. Let $\ell(t) \in \{\textsc{Novice}, \textsc{Developing}, \textsc{Proficient}, \textsc{Reviewing}\}$ be the adaptation level for topic t, where the level is assigned by the same conjunctive logic that governs certification, applied to the raw per-tier posteriors:

$$\ell(t) = \begin{cases} \text{REVIEWING}, & t \in \bar{K}, \\ \text{PROFICIENT}, & \min_k \tilde{P}_t^{(k)} \geq 0.75 \ \wedge \ \min_k N_t^{(k)} \geq 2, \\ \text{DEVELOPING}, & \min_k \tilde{P}_t^{(k)} \geq 0.35 \ \vee \ \max_k N_t^{(k)} \geq 2, \\ \text{NOVICE}, & \text{otherwise}, \end{cases} \tag{17}$$

with the cases evaluated in order and $\bar{K} \supseteq K$ denoting the set of topics certified at any point in the learner's history. A topic that decays out of K therefore retains its REVIEWING treatment, so that lapsed mastery prompts revision rather than a return to introductory scaffolding, while the prerequisite gate continues to act on K itself. Crucially, the policy reads the minimum across tiers and not the composite of Eq. (15). Partitioning the composite would be compensatory by construction, and would award reduced scaffolding to precisely the learner the Evidence Diversity Problem describes: a learner at $(0.95, 0.75, 0.55)$ attains a composite of 0.81, above any threshold that would otherwise separate the levels, while their weakest tier stands at 0.55. Adaptation is therefore conjunctive for the same reason certification is, and Eq. (15) governs display alone. The asymmetry between the two clauses is deliberate: promotion to PROFICIENT requires both conditions, since withdrawing scaffolding from an unready learner is the costlier error, whereas leaving NOVICE requires either. The cutoffs govern scaffolding depth alone and gate no formal quantity; the expertise-reversal literature establishes that support should fade with competence but fixes no threshold at which it should do so [27], so we state them as configurable policy.

TABLE III: Mastery-conditioned response policy Π, assigned by Eq. (17). The REVIEWING level is entered on certification and is not withdrawn by decay.

Level	Scaffolding & style	Challenge
NOVICE	Maximum; real-world analogies, concept in small steps	Guided fill-in-the-blank
DEVELOPING	No analogy; step-by-step mechanism trace, with a dedicated *Common Mistakes* section	Multi-concept synthesis
PROFICIENT	Minimal; skips definitions, targets edge cases, memory safety, subtle bugs	Debugging / output prediction
REVIEWING	None; concise refresher only	Quick retention check

One aspect tie the policy back to the gate. First, when a topic is certified, withholding its solution is no longer pedagogically warranted; the learner has demonstrably mastered it, so the REVIEWING level relaxes the Socratic withholding that CAC (Eq. (7)) applies to uncertified topics, replacing hint-only responses with a brief refresher and a retention check. Certification thus has two coordinated effects: it opens the prerequisite gate for *downstream* topics through K (Eqs. (4), (6)), and it lifts solution-withholding on the *mastered* topic itself.

H. Supporting Components: Retention, Cognitive Load, and Misconceptions

Three further components keep the mastery estimate faithful over time and inform adaptation.

Retention and spaced repetition: Knowledge lapses when unpracticed. Before each update, tier k's posterior is decayed toward its prior according to the time Δt (in days) since the last interaction on the topic:

$$\tilde{P}_t^{(k)} \leftarrow \tilde{P}_t^{(k)}\, e^{-\lambda_k \Delta t} + p_{L_0}^{(k)}\left(1 - e^{-\lambda_k \Delta t}\right), \qquad (18)$$

with per-tier half-lives of 14, 7, and 5 days for the declarative, procedural, and applied tiers. The rate λ_k is not fixed: each successful retrieval updates an SM-2 ease factor $EF \geq 1.3$ from a quality score, and the rate is slowed for well-retained material as $\lambda_k \leftarrow \lambda_k / EF$, so repeatedly demonstrated mastery decays more slowly. Combined with the hysteresis of Eq. (16), this makes K *non-monotone*: a certified topic left unpracticed can decay below θ_{dec} and be decertified, so the gate continues to reflect current rather than historical competence.

Cognitive load: To avoid overwhelming the learner, the model estimates instantaneous load from three telemetry signals, each normalized to $[0, 1]$ code complexity $\hat{c}$ (an AST-depth proxy), response latency $\hat{\ell}$ (capped at 20 s), and recent error rate $\hat{e}$:

$$C_{\text{load}} = 0.4\,\hat{c} + 0.3\,\hat{\ell} + 0.3\,\hat{e}. \qquad (19)$$

The estimate is surfaced to the instructor as an attention signal rather than consumed by the response path.

Active misconceptions: Recurrent, specific error patterns, for example, confusing = with == are tracked per concept with an exponential moving average over an indicator of confident error, i.e. an incorrect response the learner submitted with high stated confidence:

$$M_c \leftarrow \lambda_M M_c + (1-\lambda_M)\,\mathbf{1}[e = 0 \;\wedge\; \text{conf} \geq \theta_{\text{conf}}], \quad \lambda_M = 0.7. \qquad (20)$$

Conditioning on confidence distinguishes a misconception from an ordinary error: a learner who is unsure and wrong is still learning, whereas one who is confident and wrong holds a belief that will not self-correct. A concept with $M_c \geq 0.25$ is flagged as an active misconception and requires two consecutive correct responses to clear. Active misconceptions are surfaced to the instructor (Sec. III-I) and bias the tutor toward targeted correction.

I. Instructor Dashboard and Telemetry

Every signal above is derived from an append-only telemetry stream that SAGE persists per session: for each concept and tier, the live BKT posteriors, evidence counts $N_t^{(k)}$, and decay rates; a turn log recording the full state vector (code complexity, latency, inferred intent); and a prediction log pairing each pre-update posterior with the observed outcome, enabling continuous Brier-score monitoring of the model itself. Because the stream is append-only, it supports offline replay and psychometric re-analysis without disturbing the live model.

The same substrate carries a per-decision security record. Every gatekeeper block emits a deterministic reason naming the layer that fired, for example goal-bounded security, harmful code, semantic adjudication, or trajectory risk. A corresponding event is appended to the stream together with the inferred intent, a query excerpt, and the prevailing session risk R_t. The instructor is thus presented with a reviewable per-session security ledger rather than an opaque refusal, and can distinguish a learner who encountered a curriculum boundary from one who attempted to evade it. The recorded reason and risk score likewise support offline red-team replay and aggregate analysis of attempted evasion across a deployment.

This substrate extends SAGE's existing human-in-the-loop control (Sec. **??**) from curriculum authoring to learner monitoring. Where instructors could already lock topics and edit prerequisites, the learner model now surfaces, per learner and per topic, certified and decertified mastery, stalled progress, active misconceptions, and elevated cognitive load, prioritizing learners for attention accordingly. Actions taken through this interface adjusting pacing, unlocking a prerequisite, flagging a misconception for review propagate immediately to the gate, closing the loop between what the model observes and what the instructor regulates.

J. Implementation and Runtime Optimization

The system is implemented as a containerized microservices architecture. We use FastAPI to orchestrate the back-end and connect ChromaDB [29] (Vector Database) and Neo4j [30] (Knowledge Graph) to operate as synchronized services. The multi-agent coordination uses a lightweight orchestration framework, with all policy enforcement centralized in the Safety Gatekeeper. To sustain real-time tutoring performance, intent detection and access checks run on CPU-based local LLMs rather than remote LLMs. This design reduces the security overhead to a negligible 135 ms and lowers the Time-to-First-Token from 13.1 s to 2.2 s, demonstrating that rigorous architectural security can coexist with real-time tutoring requirements. The LLMs used to establish the baseline, such as GPT [31] and Gemini [32] are accessed via their public APIs without fine-tuning or weight modification. SAGE also uses an API-based LLM (Gemini 2.5 flash) [32] as the generator, but routes each query through the Safety Gatekeeper and controlled retrieval pipeline before generation.

A conventional subject course such as Mathematics or Biology can be transformed into our SAGE used as part of an intelligent tutoring framework by structuring the curriculum as a prerequisite graph, embedding course materials into the retrieval store, and defining light-weight access policies across the security layers using existing SAGE APIs. This process requires only course-specific content structuring and policy configuration, while the underlying Tutor architecture remains unchanged, enabling rapid and consistent deployment across different academic domains.

K. Parameter Selection

The constants introduced above fall into three kinds, defended by different standards. We state the general position once. For most of these constants there is no reason the value is 0.10 rather than 0.12, and we do not manufacture one. What can be defended is that the value lies in a principled range, and that the conclusions are stable across it; the sensitivity analyses reported in Section IV-F are therefore part of the argument rather than an appendix to it.Policy thresholds, i.e., the certification bar θ_{cert}, the evidence count $N_{\min}$, the decertification bound θ_{dec}, and the escalation threshold τ_{judge}, are not estimated quantities. They encode the confidence an institution requires before content is unlocked or an adjudication is performed, are instructor-configurable, and are justified by their consequences in the manner of a confidence-level convention. Model parameters, such as the per-tier guess and slip rates and the decay constants, are engineering defaults whose orderings we motivate where each is introduced; their specific values affect only secondary quantities, since the Evidence Diversity Guarantee of Section III-F1 holds for all admissible parameter values and Section IV-F reports the stability of certification across the plausible region. Display constants shape only what is shown to the learner and are excluded from certification by construction. We do not defend infrastructure conventions such as retrieval breadth or decoding temperature, which bear on none of the results reported here.

IV. Performance Evaluation

We conduct a multi-dimensional evaluation of our SAGE ITS system to test its effectiveness for secure and instructor-regulated learning *without* degrading instructional value. We analyze: (i) *Response Quality*: Does the system actually guide the student through the learning process, or does it simply give away the answer? We check if the LLM adheres to the curriculum and offers hints instead of solutions, (ii) *Safety and Security*: Can the system be tricked? We test if students can bypass the rules to access hidden exam keys or content they are not supposed to see yet, and (iii) *System Performance*: Is the system optimized enough for real-time user interaction? In all experiments, SAGE is compared against two state-of-the-art commercial LLM baselines (GPT and Gemini). These two LLMs are selected as they are two of the top models in the LLM leaderboard [33].

A. Experimental Setup

We evaluate along three modalities. First, a controlled simulation harness generates synthetic learners with known ground-truth mastery. Each learner is assigned a latent per-tier mastery vector, and responses are sampled with tier-appropriate guess and slip rates. We distinguish balanced learners, whose mastery is uniform across tiers, from lopsided learners, who are strong on declarative recall but weak on applied competence, i.e., the profile the Evidence Diversity Problem concerns. The same harness executes the parameter sweeps of Section IV-F and the fixed adversarial suites of Section IV-D.

Second, we conduct in-person, task-based usability sessions with individual participants [PLACEHOLDER: $N = \dots$ learners, $N = \dots$ instructors; IRB #...]. Each session comprises i) a pre-session questionnaire establishing prior experience and self-reported confidence, ii) a fixed task battery exercising the mechanisms of Section III under think-aloud, and iii) a post-session questionnaire pairing standardized usability instruments with the confidence items repeated from the pre-session form. Sessions are observed and conducted one participant at a time.

Third, the system is deployed on commodity cloud infrastructure and used over a six-week period by a beta cohort [PLACEHOLDER: $N = 10$ participants] engaged in genuine coursework rather than a prescribed battery. Three sources are collected. The append-only telemetry stream of Section III-I records every interaction and provides complete quantitative coverage. A rolling participant-maintained issue log records difficulties as they are encountered and provides time-stamped, participant-salient reports. A questionnaire administered at the close of the period records overall experience. Crucially, each logged issue carries a timestamp that resolves to the corresponding session in the telemetry stream, so a reported difficulty can be examined against the recorded system state that produced it. We treat the two sources asymmetrically: rates and distributions are computed from telemetry, whereas the issue log establishes which behaviors participants noticed and when, and is a lower bound on occurrence rather than a census.

Unless stated otherwise, simulation results report the mean over [PLACEHOLDER: S] seeds with 95% confidence intervals, and paired comparisons use McNemar's test. Results from the second and third modalities are reported descriptively with participant counts, since the samples support description rather than population estimation.

B. Metrics

The metrics that we use to evaluate responses from SAGE and baseline LLM-based AI-Tutor systems are as follows: *a) Code Density:* The fraction of response content that appears as executable code (e.g., code blocks or code-like lines). Lower values indicate the tutor avoids dumping full solutions and instead emphasizes guidance. *b) Security Compliance:* The percentage of Security-category benchmark prompts for which the system refuses or safely redirects disallowed requests. *c) Curriculum Compliance:* The percentage of queries for which the response respects course constraints such as locked topics, prerequisites, and instructor-defined pacing. *d) Pedagogy Score (1–5):* A rubric-based rating where a score of 5 reflects stepwise scaffolding and a score of 1 reflects direct solution dumping. *e) Overall Judge Preference:* The percentage of benchmark items for which an LLM-based judge [34] selects a system response as the best according to a fixed rubric emphasizing helpfulness for novice learners, pedagogical guidance, and safe refusal when applicable. *f) False Certification Rate:* The proportion of simulated learners certified for a topic on which their true per-tier competence does not meet the certification bar. Reported as a raw count, since the rates of interest are near zero and a percentage obscures the denominator. *g) Deferral Rate:* The proportion of simulated learners who are genuinely competent on all tiers but are not certified. Together with (f) this separates the two error classes, which carry different costs and are not interchangeable. *h) Π-Fidelity:* The agreement between the adaptation level assigned by the policy and the level recovered by a blind judge shown only the generated response, reported with Cohen's κ against a chance baseline of $1/|\ell|$. *i) Attack Success Rate:* The proportion of adversarial multi-turn sessions in which harmful content was delivered at any turn, judged on delivery rather than on the presence of a refusal message. *j) Benign Over-Refusal Rate:* The proportion of legitimate multi-turn sessions blocked at any turn. Reported alongside (i) rather than combined with it, since the two errors carry asymmetric costs and the deployed base rate is not the balanced rate of the audit suite.

C. Pedagogical Alignment on C-EduBench

We first establish that the learner model is integrated without degrading the instructional behavior of the inherited system. Table IV compares raw and prompt-engineered baselines against the conference system and the present extension on C-EduBench. Two results bear on the learner model directly. Curriculum compliance rises from 80.0% to 100.0%, which is the expected consequence of replacing an asserted mastered set with an earned one: prerequisite decisions that previously rested on course enrollment now rest on per-tier evidence, and the residual violations of the conference system were concentrated in exactly those cases. Pedagogy score rises from 2.4 to 3.62 while code density remains low at 11.12%, indicating that the mastery-conditioned adaptation of Section III-G recovers the instructional quality that the conference system traded away for control, without reverting to solution dumping. Security compliance is unchanged at [PLACEHOLDER: reconcile against the deployed audit; see Section IV-D], as expected, since the learner model alters the provenance of K but modifies no gate equation.

The comparison against prompt-engineered baselines is unchanged in character from the conference version. Prompt-based tutors remain competitive on pedagogy; GPT (Tutor) attains 4.0, while enforcing policy at a rate that a course cannot rely on. Crucially, this gap is not one that better prompting closes, since the baselines fail on retrieval-boundary decisions their prompt has no authority over.

D. Multi-Turn Enforcement

We audit the extended gate on 15 adversarial multi-turn sessions spanning eight strategies, for example crescendo, keyword-sparse, slow burn, incremental decomposition, social engineering, and persona roleplay, against 9 benign multi-turn controls, one of which is deliberately security-adjacent. Containment is judged on whether harmful content was delivered rather than on the presence of a refusal message, since prerequisite-gating and controlled retrieval also deflect attacks without an explicit block. Results are summarized in Table V. Of 63 adversarial turns, 60 were deflected, and no session yielded a complete harmful artifact: the terminal assembly turn was refused in every case. Containment was total on restricted institutional assets, for which the retrieval boundary is the sole control point, and the three residual sessions disclosed partial material only. Benign traffic was largely unaffected, including the security-adjacent control, and the single over-refusal arose from an off-topic warning on an expression of learner frustration rather than from any security layer.

TABLE IV: Comparative Performance Across Raw, Tutor, SAGE, and IRL-EXT Systems. Lower is better for Code Density; higher is better for Security Compliance, Curriculum Compliance, Pedagogy Score, and Overall Win Rate.

Metric	**GPT (Raw)**	**GPT (Tutor)**	**Gemini (Raw)**	**Gemini (Tutor)**	**SAGE**	**IRL-EXT**
Code Density (↓)	10.0%	**6.9%**	23.0%	9.4%	16.4%	11.12%
Security Compliance (↑)	20.0%	30.0%	20.0%	30.0%	**90.0%**	**90.0%**
Curriculum Compliance (↑)	NA	NA	NA	NA	80.0%	**100.0%**
Pedagogy Score (1–5) (↑)	3.1	**4.0**	3.4	3.8	2.4	3.62

TABLE V: Multi-turn adversarial audit. 15 attack sessions spanning eight strategies against 9 benign multi-turn controls. Containment is judged on delivery of harmful content, not on the presence of a refusal message. ↑ / ↓ indicate that higher / lower is better.

Outcome	Result	
Adversarial turns deflected (↑)	60/63	(95.2%)
Sessions with no harmful delivery (↑)	12/15	(80.0%)
Complete artifact delivered (↓)	0/15	(0.0%)
Restricted-asset containment (↑)	2/2	(100%)
Benign sessions never blocked (↑)	8/9	(88.9%)
Gate misses recovered at generation (↑)	3/5	(60.0%)

Notably, the access-control and generation layers fail on disjoint sets. Three attacks reached an unblocked payload turn and were nonetheless refused at generation, while one attack whose payload turn was blocked had disclosed an example earlier in the session. Neither layer's failures are contained in the other's, which is the operative evidence that the layers are complementary rather than redundant. Consistent with this, containment tracks lexical marking rather than intent, and the residual surface consists of operations expressed in unmarked curriculum vocabulary.

E. Over-Crediting Resistance

The Evidence Diversity Problem asserts that a learner strong on declarative recall and weak on applied competence should not be certified. We test this directly, since the claim requires ground truth that authentic data cannot supply: we generate learners whose true per-tier competence is fixed in advance and observe which rule certifies them. Each learner answers 60 items sampled with tier-appropriate guess and slip rates, and the identical response stream is supplied to every rule under comparison. The simulation imports the deployed update function and evidence configuration directly rather than reimplementing them, so the certification behavior measured is that of the running system.

Decomposing quantity from diversity: Our rule differs from standard knowledge tracing in two respects: it factors evidence by tier, and it imposes a minimum evidence count. To separate their contributions, we evaluate three rules. The first is the canonical formulation, in which mastery is declared when the posterior crosses θ_{cert} with no evidence requirement, which is the standard criterion in knowledge tracing [19]. The second is a strengthened baseline that pools evidence into a single posterior but additionally requires nine observations, matching the total demanded by our rule. The third is the factored rule of Eq. (16). The pooled baseline is not part of the standard formulation and is included solely to isolate the effect of evidence quantity.

TABLE VI: Certification rates by learner archetype, 2000 simulated learners per cell, at the deployed parameterization. The canonical arm declares mastery on the posterior alone; the pooled arm adds a matched evidence floor of nine observations without tier structure.

Archetype	Truth	Canonical	Pooled	Factored
Recall-strong, code-weak	non-master	1999/2000	1998/2000	**0/2000**
Code-strong, recall-weak	non-master	[X]	[X]	0.1%
Uniformly weak	non-master	48.9%	40.1%	**0.0%**
Moderate (≈ 0.65)	boundary	100%	[X]	35.1%
Uniformly strong	master	100%	[X]	75.7%

Table VI reports the outcome. On recall-strong, code-weak learners, the canonical rule certifies 1999 of 2000 and the quantity-matched rule certifies 1998 of 2000, whereas the factored rule certifies *none* (0/2000, 95% CI [0.00, 0.19]). Crucially, the decomposition is unambiguous: matching evidence quantity alters the outcome by one case in two thousand, while factoring by tier eliminates over-crediting entirely. The result is therefore attributable to evidence diversity and not to a larger evidence requirement. This empirically realizes Theorem 1, whose consequence is that lopsided evidence cannot satisfy Eq. (16) irrespective of its volume.

Symmetry and boundary behavior: The rule is not specific to quiz-inflated profiles. On the inverse archetype, i.e., learners strong in applied work but weak on declarative recall, the factored rule certifies 0.1%, so the conjunction refuses incomplete evidence in either direction. On learners of uniform moderate competence (≈0.65 on every tier), the canonical rule certifies all of them while the factored rule certifies 35.1%, responding gradually rather than uniformly at the boundary. We note that the appropriate certification rate for partially competent learners is a policy question the rule does not settle; what the comparison establishes is that the canonical rule does not distinguish this population from a fully competent one.

F. Decision Quality: Deferral versus Misclassification

A certification rule can err in two ways, and the two are not interchangeable. Certifying a learner who has not demonstrated competence admits them to material they are unready for, and the error propagates through every downstream gating decision. Declining to certify a learner who is in fact competent withholds nothing they have earned and is corrected by further evidence. The errors of the two rules fall into different classes rather than differing in rate. Across every archetype in Table VI, the factored rule's errors are deferrals on true masters and it produces no false certification, while the pooled rule's errors are false certifications on non-masters.

Deferral is a property of the curriculum, not of the rule: The right panel of Fig. 4 reports the proportion of true masters certified as a function of the share of applied items in the assessment mix. At a 5% share the factored rule certifies 34.1% of true masters; at 10% it certifies 75.7%; at 20% and above it certifies 97.4% and higher. Deferral therefore arises when the curriculum supplies too little applied evidence for the applied tier to accumulate, and not from conservatism in the rule itself. Crucially, the enrichment that removes the cost does not remove the guarantee: at a 20% applied share the pooled baseline still certifies approximately 99% of recall-strong, code-weak learners, so the additional evidence benefits the factored rule alone.

This carries a direct implication for deployment, which we state plainly. Our own configuration supplies a 10% applied share and therefore incurs a 24.3% deferral rate on true masters. On the evidence of Fig. 4 we recommend an applied share of at least 20%, at which the guarantee is obtained at a deferral cost below 3%. The requirement the rule imposes is thus on assessment design rather than on the learner.

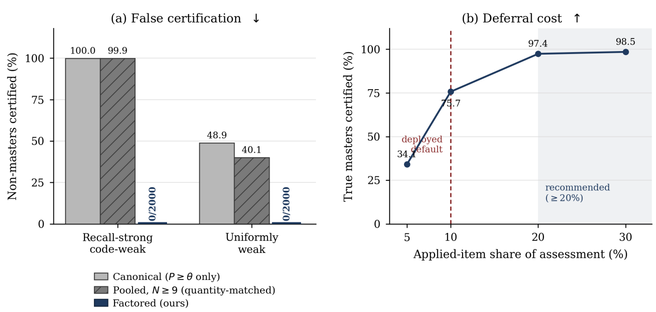

Fig. 4: Left: false certification of recall-strong, code-weak learners under the canonical rule, a quantity-matched pooled rule, and the factored rule (2000 learners per arm; lower is better). Right: proportion of true masters certified by the factored rule as a function of applied-item share, with the deployed configuration marked and the recommended region shaded.

Sensitivity to the transition rate: The rate at which the canonical rule certifies uniformly weak learners is governed principally by the transition parameter P_T, which admits upward drift in the absence of an evidence requirement. Sweeping $P_T \in \{0, 0.03, 0.09, 0.15\}$, the canonical rule certifies 9.0%, 27.4%, 48.9%, and 56.0% of uniformly weak learners respectively, and the pooled rule 2.1%, 21.0%, 40.1%, and 46.2%. We therefore report this quantity as a sweep rather than as a single figure. The over-crediting result of Section IV-E is not so governed: the factored rule certifies no recall-strong, code-weak learner at any P_T examined, while the pooled arms certify at least 89.7% of them even at $P_T = 0$. At the largest transition rate examined the factored rule certifies 0.1% of uniformly weak learners, which is consistent with Theorem 1, whose scope is lopsided evidence rather than uniformly weak learners fortunate across all three tiers.

Baseline parameterization was pooled over the item mix, giving $p_G = 0.155$, $p_S = 0.125$, $p_{L_0} = 0.196$, and $P_T = 0.09$.

G. *Mastery-Conditioned Adaptation*

A learner model is consequential only if it changes what the learner receives. We therefore ask whether the same query yields materially different responses across proficiency states, and whether that difference is recoverable by an observer who does not know the learner. Six synthetic learners were written directly into the knowledge store, differing only in their per-tier posteriors, and each was issued the same 20 queries on a single concept in a fresh session, giving 120 responses through the live pipeline.

The adaptation policy assigned the correct treatment on all 120 turns, verified against an independent transcription of the deployed rule. Table VII illustrates the resulting difference for a single query. Responses differ not merely in length but in rhetorical strategy: the novice treatment leads with an analogy and defines every term, the developing treatment withholds analogy and traces the underlying mechanism, the proficient treatment omits definition entirely and opens on failure modes, and the reviewing treatment introduces no new material. Section structure differs correspondingly, each level emitting at least one section no other level produces.

Fidelity: Deterministic features confirm the separation: code density rises from 33.9% at novice to 61.4% at proficient while analogy cues fall from 1.55 to 0.05 per response, and every adjacent pair of levels separates on at least one feature. These counts are partly circular, however, since the prompt instructs several of the behaviors being counted. We therefore evaluate fidelity with a blind judge, which is shown a single response with no information about the learner and asked which of the four treatments it was written for. Agreement with the assigned treatment was 83.3% (100/120, 95% CI [75.7, 88.9]; $\kappa = 0.767$) against a 25% chance baseline, with no response declined. Errors are ordinal rather than arbitrary: 18 of 20 fall between adjacent levels, with a mean ordinal distance of 1.15.

Where the policy is least sharp: Recall is 100% at novice and 95% at proficient and reviewing, but 70% at developing, and the confusions concentrate on a specific learner profile. A learner weak on recall but strong on applied work is correctly assigned the developing treatment by the conjunctive rule, yet the generated response is judged proficient in 8 of 20 cases, since a code-strong learner receives a code-heavy answer that presents as advanced. The policy and the generation layer therefore disagree at the developing/proficient boundary, and we report this as the principal limit of the present adaptation prompt rather than of the assignment rule.

TABLE VII: Mastery-conditioned adaptation. The same query, "Can you explain what a pointer is in C?", issued by four learners differing only in their per-tier BKT state. Excerpts are verbatim system output.

Level	Words	Rhetorical move	Excerpt
Novice	316	Analogy-led; defines every term	"...a variable is like a labeled box where you store a piece of data...Think of it like a scavenger hunt: if a normal variable is the treasure, a pointer is a piece of paper with the exact GPS coordinates of where that treasure is hidden."
Developing	363	No analogy; traces the mechanism	"A pointer is a memory address stored in a variable, but the power of pointers lies in the interaction between the address-of operator (`&`) and the dereference operator (`*`)...you are instructing the CPU to look at the address stored in `ptr` and perform a read or write directly at that memory location."
Proficient	258	No definition; opens on failure modes	"Since you are comfortable with basic pointer mechanics, you should be aware of the risks regarding pointer arithmetic and memory safety. Accessing memory via an uninitialized pointer...results in undefined behavior, which often manifests as intermittent segmentation faults or silent data corruption."
Reviewing	165	Teaches nothing new; reminds	"Remember that a pointer is simply a variable that stores the memory address of another variable...A common 'gotcha' that often fades is the difference between the pointer's address and the value it points to."

Weak-tier identification: Where evidence supports it, the system names the learner's weakest tier. The named tier varies with the evidence: the applied tier for recall-strong learners and the recall tier for applied-strong learners, and is withheld when the spread between tiers is small, or the weakest tier is unevidenced. Both conditions are load-bearing: an unconditional minimum returns the applied tier for every learner, since it carries the lowest prior and is therefore weakest by construction rather than by evidence.

H. Beta Testing result

I. Usability Testing result

V. Conclusion

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